Laravel Portfolio Project Report

1) Laravel Installation Procedure

1. Install prerequisites
2. PHP (8.1+ recommended)
3. Composer
4. MySQL (if DB is needed)
5. Node.js + npm (for frontend assets)
6. Create a Laravel project using the command—**composer create-project "laravel/laravel:^13.0" myportfolio**
7. cd myportfolio
   Configure environment
8. Copy env file:
9. cp .env.example .env
10. Generate app key:
11. php artisan key:generate
12. Set database credentials (optional if using DB)
    Update .env:
    DB_DATABASE=...
    DB_USERNAME=...
    DB_PASSWORD=...

Run migrations (if database tables are required)

13. php artisan migrate
    Start the development server
14. php artisan serve
15. Open browser: http://127.0.0.1:8000

2) Bootstrap Integration Process in Laravel

Method used:
Import CSS
Import Bootstrap in /public/css/custom.css:
Import Bootstrap in /public/css/style.css:

Import JS
Import Bootstrap JS in public/js/custom.js:
Import Bootstrap CSS in public/js/script.js:

Import Images
For the images I imported online using links to link with <img src=""/> tag

Integrating CSS and JS in Laravel Pages

I integrated them using asset example:

```
<script src="{{ asset('js/scripts.js') }}"></script>
 <script src="{{ asset('js/custom.js') }}"></script>
```

```
<link href="{{ asset('css/styles.css') }}" rel="stylesheet" />
 <link href="{{ asset('css/custom.css') }}" rel="stylesheet" />
```

3) Project Overview

This project is a Laravel web application with four main pages:

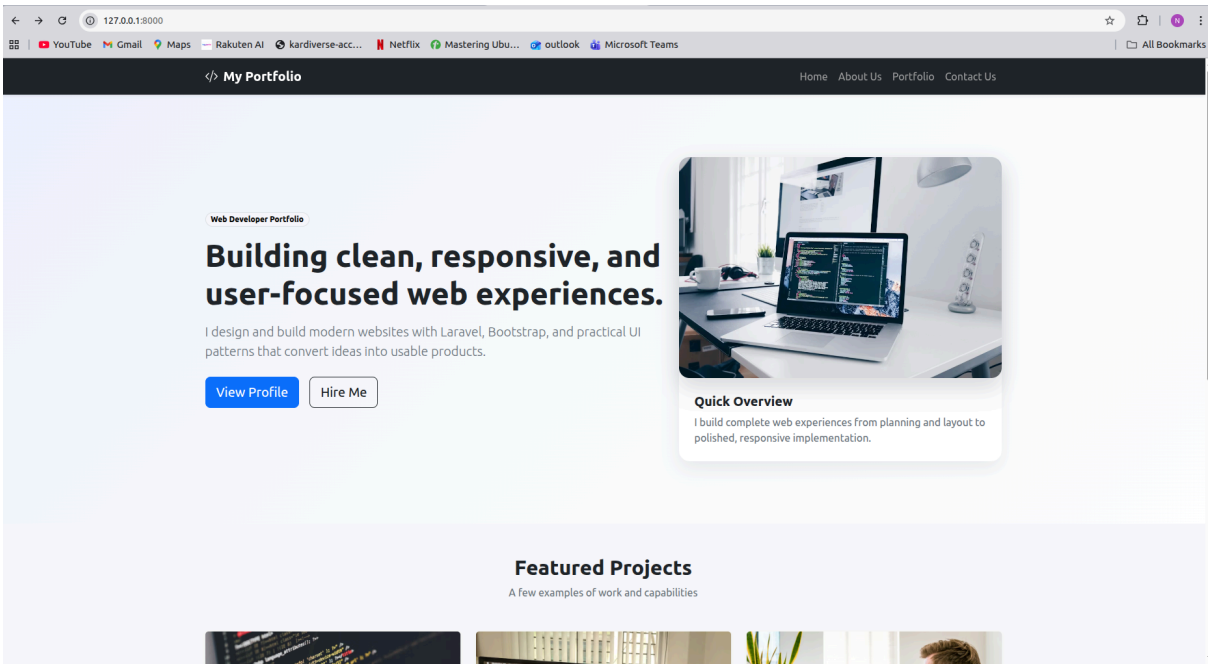
- Home
- Portfolio
- Contact Us
- About Us

The aim is to demonstrate basic Laravel routing, Blade templating, and Bootstrap-based responsive UI design.

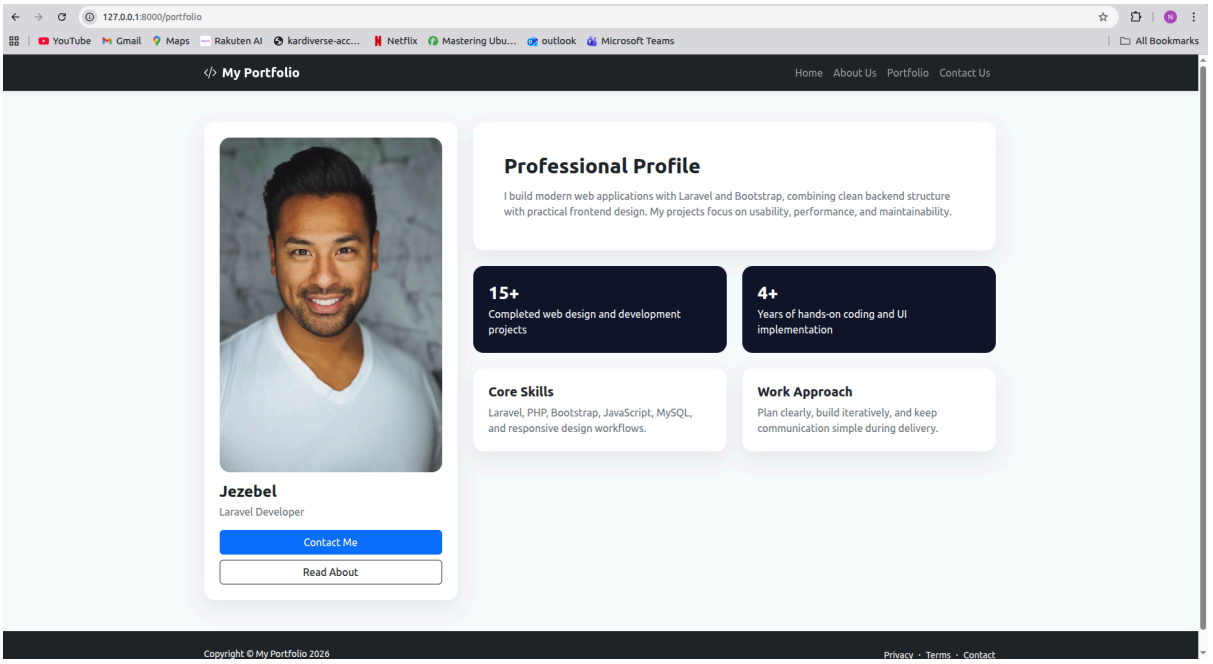
4) Application Screenshots (Browser Output)

4 screenshots in this section (one per page).

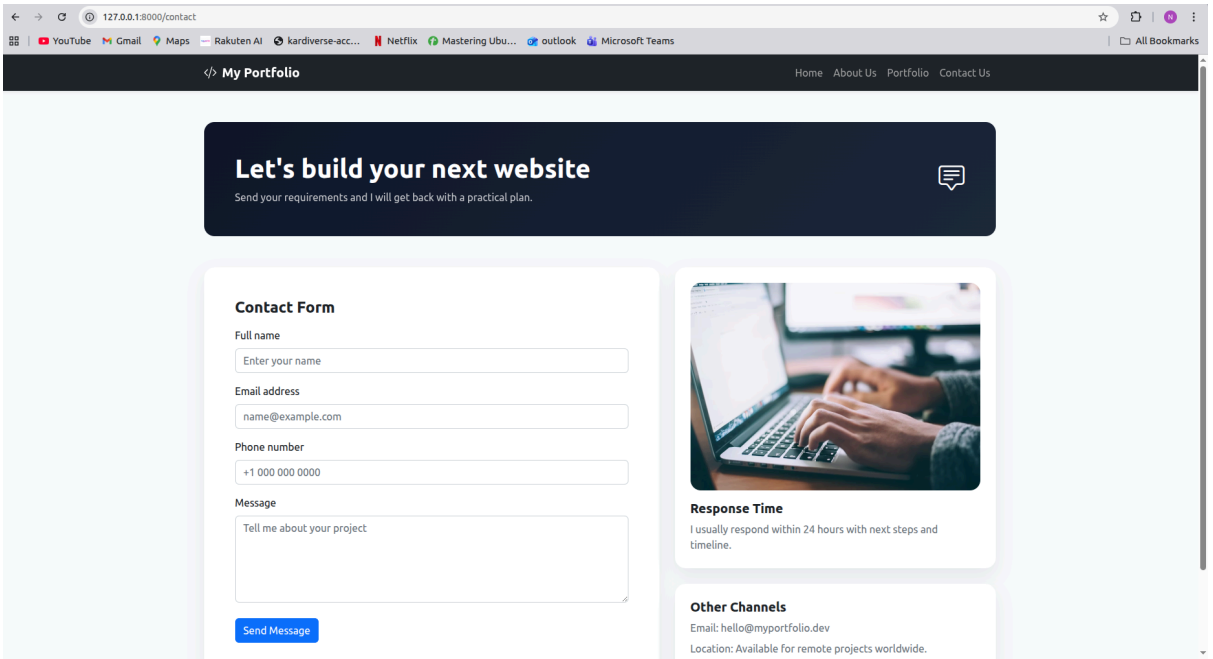
Home Page – Shows landing section and navigation bar.



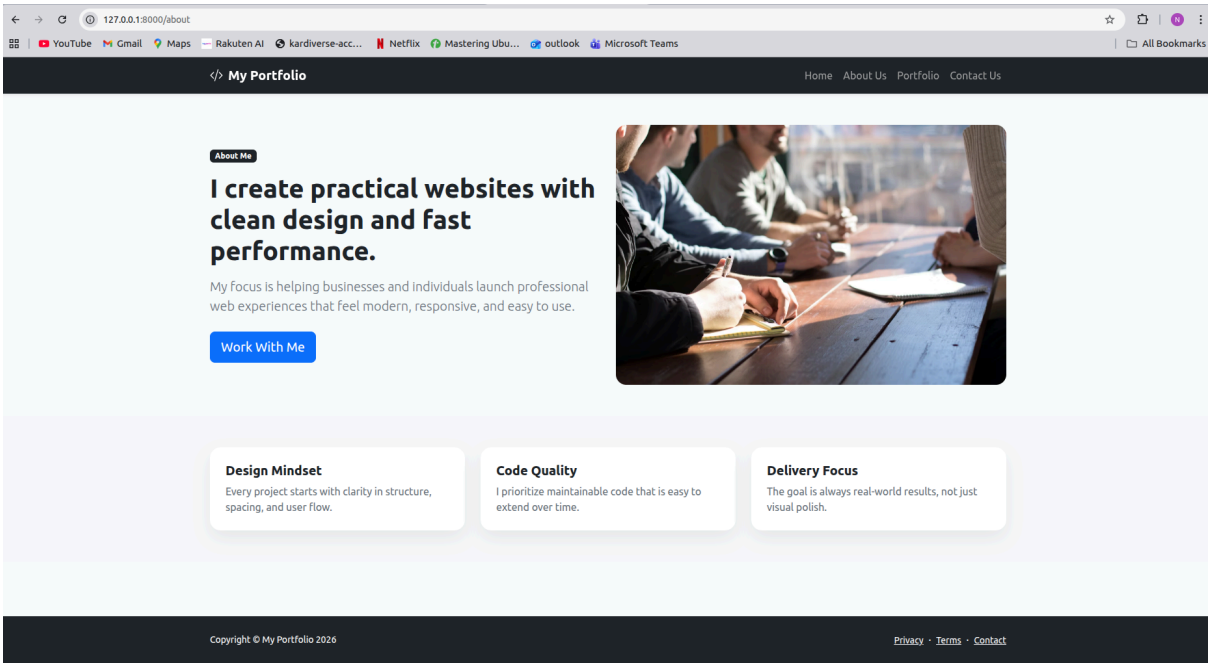
Portfolio Page – Displays project/work samples in card layout.



Contact Us Page – Contains contact form and user input fields.



About Us Page – Describes profile background.



5) Challenges Encountered During Development

Common issues:

- **Missing dependencies**
  composer/npm packages not installed correctly.
  Fix: run composer install and npm install.

- **Styles/scripts not loading.**
  Fix: Did not integrate well in the Blade template.
  Fix: Ensured Bootstrap import and inclusion are correct.
  Permission issues

- **Route names/URLs mismatch. Route not found (404**
  Fix: verify routes in routes/web.php.

- **Wrong PHP version or missing extensions.**
  Fix: check with php -v and install required extensions (mbstring, openssl, pdo, etc.).
  Port conflicts

- **php artisan serve fails due to the used port.**
  Fix: use another port.
  php artisan serve --port=8000

6) Conclusion

The Laravel portfolio application successfully demonstrates multi-page routing and responsive UI integration using Bootstrap. The project provides a practical foundation for building dynamic web applications and highlights important setup, build, and debugging workflows in Laravel development.